APEM journal

Advances in Production Engineering & Management **ISSN 1854-6250**
Volume 16 | Number 3 | September 2021 | pp 285–296 Journal home: apem-journal.org
https://doi.org/10.14743/apem2021.3.400 Original scientific paper

Using the gradient boosting decision tree (GBDT) algorithm for a train delay prediction model considering the delay propagation feature

Zhang, Y.D.[a,*], Liao, L.[a], Yu, Q.[a], Ma, W.G.[a], Li, K.H.[b]

[a]School of Information Science and Technology, Southwest Jiaotong University, Chengdu, P.R. China
[b]School of management, Xihua University, Chengdu, P.R. China

A B S T R A C T

Accurate prediction of train delay is an important basis for the intelligent adjustment of train operation plans. This paper proposes a train delay prediction model that considers the delay propagation feature. The model consists of two parts. The first part is the extraction of delay propagation feature. The best delay classification scheme is determined through the clustering method of delay types for historical data based on the density-based spatial clustering of applications with noise algorithm (DBSCAN), and combining the best delay classification scheme and the k-nearest neighbor (KNN) algorithm to design the classification method of delay type for online data. The delay propagation factor is used to quantify the delay propagation relationship, and on this basis, the horizontal and vertical delay propagation feature are constructed. The second part is the delay prediction, which takes the train operation status feature and delay propagation feature as input feature, and use the gradient boosting decision tree (GBDT) algorithm to complete the prediction. The model was tested and simulated using the actual train operation data, and compared with random forest (RF), support vector regression (SVR) and multilayer perceptron (MLP). The results show that considering the delay propagation feature in the train delay prediction model can further improve the accuracy of train delay prediction. The delay prediction model proposed in this paper can provide a theoretical basis for the intelligentization of railway dispatching, enabling dispatchers to control delays more reasonably, and improve the quality of railway transportation services.

A R T I C L E   I N F O

Keywords:
Train delay prediction;
Actual train operation data;
Delay type identification;
Delay propagation feature extraction;
Density-based spatial clustering of applications with noise (DBSCAN);
k-nearest neighbor (KNN);
Gradient boosting decision tree (GBDT);
Random forest (RF);
Support vector regression (SVR);
Multilayer perceptron (MLP)

**Corresponding author:*
ydzhang@swjtu.edu.cn
(Zhang, Y.D.)

Article history:
Received 24 July 2021
Revised 25 October 2021
Accepted 28 October 2021

1. Introduction

With the development of railway network and the growth of passenger travel demand, the utilization rate of railway lines is getting higher and higher. Under the premise of ensuring the safety of train operation, ensuring the punctuality is the key of railway transportation to improve the quality of service. Once the train is delayed, the dispatchers must use dispatch adjustment methods reasonably and scientifically [1]. The accurate prediction of train delays can assist dispatchers to make scientific decisions, and even realize the intelligent dynamic adjustment of train operation plans.